

In this paper, we observe that typical manifolds,

We promise that: equivalent relation on X generated by: condition means the smallest equivalent relation on X satisfying the condition.

Rigorously, $\sim := \bigcap R_1 \times R_2$
 $R_1 \times R_2 \in \mathcal{X} \times \mathcal{X}$ is eq-rel
 $R_1 \times R_2$ satisfies the condition

And, $I = [0, 1]$, the subspace of the euclidean space $\mathbb{R}$.

Cylinder

Def.

Define an equivalence relation $\sim$ on $I \times I$ generated by: $\forall y \in I, (0, y) \sim (1, y)$
The quotient space $I^2/\sim$ is called the Cylinder.

Prop. The cylinder $I^2/\sim$ is Homeomorphic to $S^1 \times I \subseteq \mathbb{C} \times \mathbb{R}$.

Proof. Define a map

$$f: I^2 \rightarrow S^1 \times I: (x, y) \mapsto (e^{2\pi i x}, y)$$

$$\begin{array}{ccc} I^2 & & f \\ & \searrow & \\ \pi \downarrow & & S^1 \times I \\ I^2/\sim & \xrightarrow{\tilde{f}} & \end{array}$$

Then, f is continuous onto closed map, because

1. Continuous

$(x, y) \mapsto e^{2\pi i x}$ and $(x, y) \mapsto y$ are continuous, so f is also continuous.

2. Onto

Given $(e^{i\theta}, b) \in S^1 \times I$ where $\theta \in [0, 2\pi]$, take $x = \frac{\theta}{2\pi}$ and $y = b$, then clear.

3. Closed

Given closed set $C \subseteq I^2$, C is compact since I^2 is compact.

So, the continuous image $f[C]$ is also compact in $S^1 \times I$,

therefore $f[C]$ is closed, by $S^1 \times I$ is Hausdorff.

Moreover, $f(x_1, y_1) = f(x_2, y_2) \Leftrightarrow e^{2\pi i x_1} = e^{2\pi i x_2}$ and $y_1 = y_2$
($\Leftrightarrow e^{2\pi i(x_1 - x_2)} = 1$)

$$\Leftrightarrow \left(x_1 = x_2 \text{ or } \{x_1, x_2\} = \{0, 1\} \right) \text{ and } y_1 = y_2$$

$$\Leftrightarrow (x_1, y_1) \sim (x_2, y_2)$$

Therefore, f induces a Homeomorphism

$$\tilde{f}: I^2/\sim \rightarrow S^1 \times I: [(x, y)] \mapsto f(x, y).$$

It is injective and well-defined by $(x_1, y_1) \sim (x_2, y_2) \Leftrightarrow f(x_1, y_1) = f(x_2, y_2)$

And surjective is given by the f is surjective.

Finally, the continuity is given by: $\tilde{f} \circ \pi = f$ is continuous, and,

given closed set $C \subseteq I^2/\sim$, $\pi^{-1}[C]$ is closed, so the ontoness of π gives

$\tilde{f}[C] = \tilde{f}[\pi[\pi^{-1}[C]]] = f[\pi^{-1}[C]]$ is closed, since f is closed map. $\square$

Torus

Def.

Define an equivalence relation $\sim$ on $I \times I$ generated by:

$$\begin{cases} \forall x, y \in I, (x, 0) \sim (x, 1) \text{ and } (0, y) \sim (1, y) \\ (0, 0) \sim (1, 0) \sim (0, 1) \sim (1, 1) \end{cases}$$

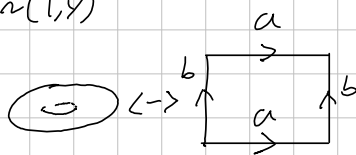
The quotient space $I^2/\sim$ is called a Torus.

Prop. The Torus $I^2/\sim$ is Homeomorphic to $S^1 \times S^1 \subseteq \mathbb{C}$.

Proof. Define a map

$$f: I^2 \rightarrow S^2: (x, y) \mapsto (e^{2\pi i \cdot x}, e^{2\pi i \cdot y})$$

Then, f is continuous onto closed map,



$$\langle a, b \mid aba^{-1}b^{-1} = e \rangle$$

$$\cong \mathbb{Z} \times \mathbb{Z}$$

$$I^2/\sim \cong S^1 \times S^1 \text{ gives}$$

$$\pi_1(I^2/\sim) \cong \pi_1(S^1 \times S^1)$$

$$\cong \pi_1(S^1) \times \pi_1(S^1)$$

Projective space.

Def. Define an equivalence relation on $\mathbb{R}^{n+1} \setminus \{0\}$ generated by:

$$x \sim y \Leftrightarrow \exists t \in \mathbb{R} \text{ such that } tx = y.$$

The quotient space $\mathbb{RP}^n \stackrel{\text{def}}{=} (\mathbb{R}^{n+1} \setminus \{0\}) / \sim$ is called a Real Projective space.

klein bottle.

Möbius band.